AcademyCloudFoundation...
Presentation

Make notes Very detailed (full sentences, no short phrases)
✅ Including all important theoretical points
✅ Including procedural steps wherever steps are given
✅ Formatted cleanly for exam
make sure to include each and every imp point.

Below are **very detailed, full-sentence exam-oriented notes for Module 5: Networking and Content Delivery**, prepared carefully from the PPT content AcademyCloudFoundations_Module_...
(All points are expanded into complete sentences, with procedures included wherever implied.)

📘 MODULE 5 – NETWORKING AND CONTENT DELIVERY (DETAILED EXAM NOTES)

🌐 SECTION 1: NETWORKING BASICS

What is a Network?

A network is a collection of interconnected devices that communicate with each other using standardized protocols in order to exchange data efficiently and reliably.

📍 IP Addresses

An IP address is a unique numerical identifier assigned to each device connected to a network so that it can be located and communicated with.

IPv4 Address:

- IPv4 is a **32-bit address**

- It is written in **four groups of numbers**
- Each group ranges from **0 to 255**
- Example: `192.0.2.0`

IPv6 Address:

- IPv6 is a **128-bit address**
- It is written in **eight hexadecimal groups**
- Each group ranges from **0 to FFFF**
- Example: `2600:1f18:22ba:8c00:ba86:a05e:a5ba:00FF`

📐 Classless Inter-Domain Routing (CIDR)

CIDR is a method for allocating IP addresses and routing traffic efficiently by defining address ranges using suffix notation such as `/16` or `/28`.

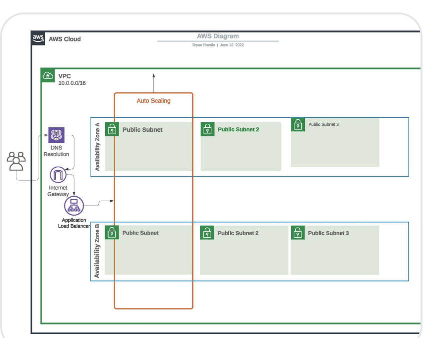
It allows:

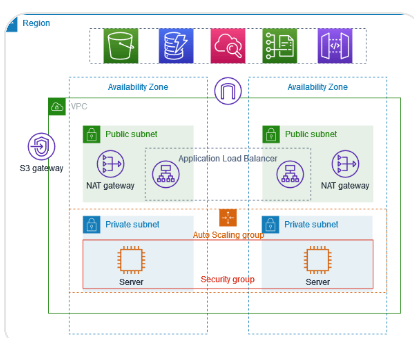
- Flexible IP address allocation
- Reduction in wasted IP addresses
- Efficient routing

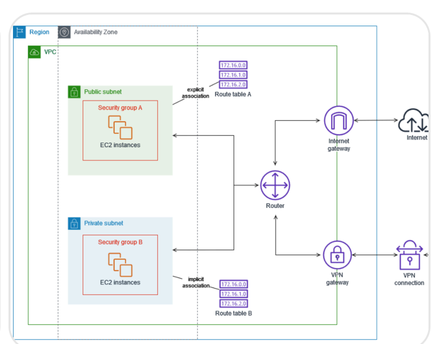
📊 OSI (Open Systems Interconnection) Model

The OSI model is a conceptual framework with **seven layers** that explain how data flows across a network, from physical transmission to application interaction.

☁️ SECTION 2: AMAZON VPC (VIRTUAL PRIVATE CLOUD)







What is Amazon VPC?

Amazon VPC allows users to create a **logically isolated virtual network** within the AWS Cloud where AWS resources can be launched securely.

It provides full control over:

- IP address ranges
- Subnet creation
- Route tables
- Gateways
- Network security layers

📦 VPCs and Subnets

VPC Characteristics:

- A VPC is isolated from other VPCs
- It belongs to **one AWS Region**
- It can span **multiple Availability Zones**

Subnet Characteristics:

- A subnet is a subdivision of a VPC's IP range
- Each subnet exists in **one Availability Zone only**
- Subnets are classified as:
    - **Public subnets** (connected to internet)
    - **Private subnets** (not directly connected)

📍 IP Addressing in VPC

When creating a VPC:

- You must assign an **IPv4 CIDR block**
- The CIDR block cannot be changed later
- Maximum size: **/16**
- Minimum size: **/28**
- Subnet CIDR blocks must not overlap
- IPv6 is also supported

🚫 Reserved IP Addresses

In every subnet:

- AWS reserves **5 IP addresses**
- These are used for:
  - Network address
  - VPC router
  - DNS services
  - Future use

Example:

- VPC CIDR: `10.0.0.0/16` → 65,536 total IPs
- Divided into 4 subnets
- Each subnet gets 256 IPs
- Only **251 IPs are usable**

🌍 Public IP Address Types

Public IPv4 Address:

- Assigned automatically at subnet level
- Or manually via Elastic IP

Elastic IP:

- Static public IP
- Associated with AWS account
- Can be remapped anytime
- May incur additional cost

🔌 Elastic Network Interface (ENI)

An ENI is a virtual network card that:

- Can be attached to an EC2 instance
- Can be detached and attached to another instance
- Retains IP addresses and attributes

Every instance has a **default ENI** with a private IP.

🧭 Route Tables

A route table controls how traffic flows within subnets.

Each route contains:

- Destination address
- Target (gateway, instance, etc.)

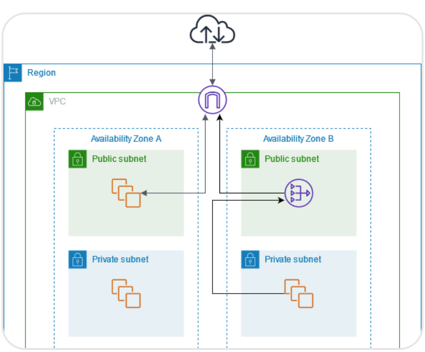
Important rules:

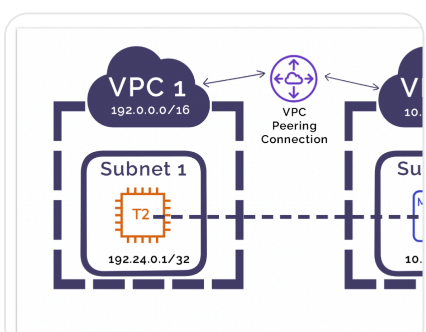
- Every subnet must be associated with one route table
- Every route table has a default **local route**
- The local route allows internal VPC communication
- The local route cannot be deleted

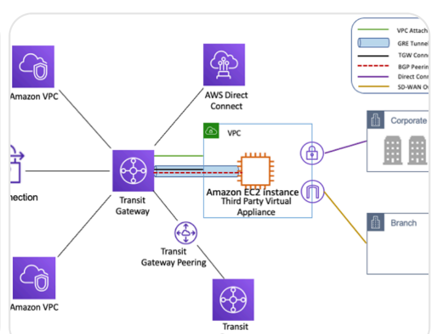
✅ Section 2 Summary

- A VPC is an isolated network in AWS
- It belongs to one Region
- It uses CIDR blocks
- It contains subnets in AZs
- Route tables control traffic

🔁 SECTION 3: VPC NETWORKING OPTIONS







Internet Gateway

An internet gateway allows public subnets to communicate with the internet.

NAT Gateway

A NAT gateway enables private subnet instances to access the internet for updates while remaining unreachable from outside.

VPC Sharing

VPC sharing allows multiple AWS accounts within the same organization to use shared subnets.

Key points:

- The owner creates the VPC
- Participants deploy resources
- Participants cannot modify others' resources

VPC Peering

A VPC peering connection connects two VPCs privately.

Rules:

- CIDR blocks must not overlap
- Transitive peering is not allowed
- Only one peering connection per pair

AWS Site-to-Site VPN

Creates encrypted tunnels between on-premises networks and AWS VPC.

AWS Direct Connect

Provides a dedicated physical connection from on-premises data centers to AWS for high speed and reliability.

VPC Endpoints

Two types:

- Interface endpoints (PrivateLink)
- Gateway endpoints (S3 and DynamoDB)

They allow private access without internet.

AWS Transit Gateway

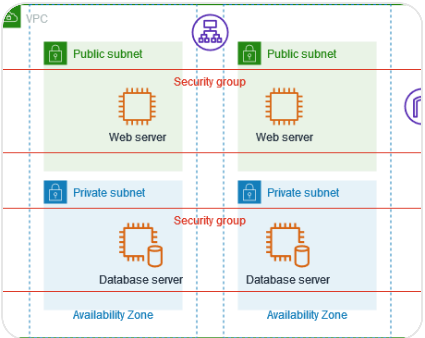
Acts as a central hub connecting:

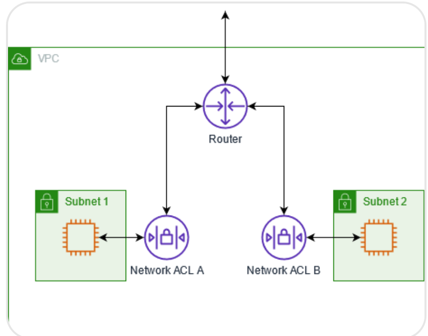
- Multiple VPCs
- VPNs
- Direct Connect

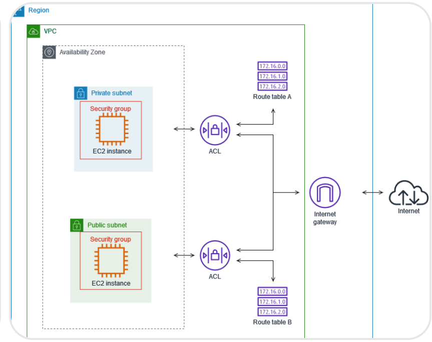
✅ Section 3 Summary

- Internet Gateway
- NAT Gateway
- VPC Peering
- VPC Sharing
- VPN
- Direct Connect
- Transit Gateway
- VPC Endpoints

🔐 SECTION 4: VPC SECURITY







🔒 Security Groups

Security groups act as **instance-level firewalls**.

Features:

- Control inbound and outbound traffic
- Allow rules only (no deny rules)
- Are stateful (return traffic allowed automatically)
- Default: deny inbound, allow outbound

🚧 Network ACLs

Network ACLs act as **subnet-level firewalls**.

Features:

- Have allow and deny rules
- Are stateless (responses must be explicitly allowed)
- Rules are evaluated in numerical order
- Default ACL allows all traffic
- Custom ACL denies all until rules added

⚖️ Security Groups vs Network ACLs

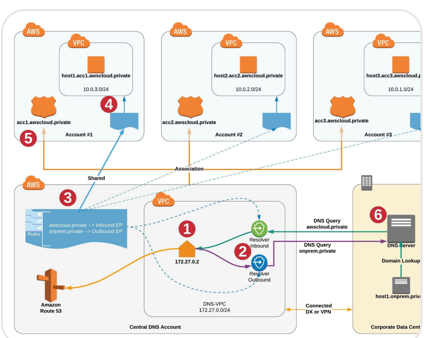
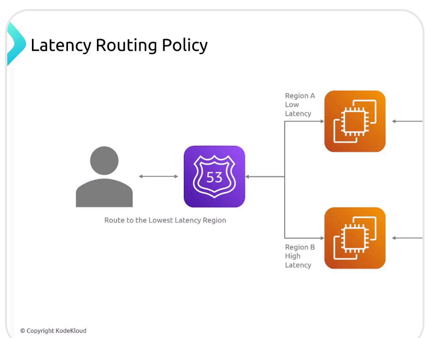
Feature	Security Group	Network ACL
Level	Instance	Subnet

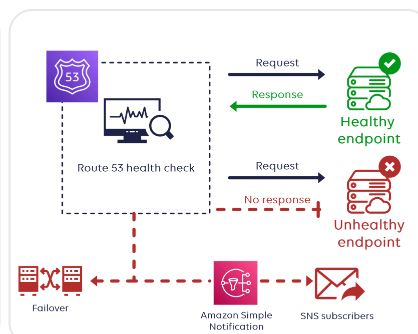
Feature	Security Group	Network ACL
Rules	Allow only	Allow + Deny
State	Stateful	Stateless
Order	All evaluated	Number order

✅ Section 4 Summary

- Isolate subnets
- Use gateways wisely
- Apply layered security
- Use security groups and ACLs

🌍 SECTION 5: AMAZON ROUTE 53



What is Route 53?

Amazon Route 53 is a scalable DNS service that:

- Converts domain names to IP addresses
- Supports IPv4 and IPv6
- Routes traffic intelligently
- Performs health checks
- Registers domains

Routing Policies Supported:

1. Simple routing
2. Weighted routing
3. Latency-based routing
4. Geolocation routing
5. Geoproximity routing
6. Failover routing
7. Multivalue routing

DNS Failover

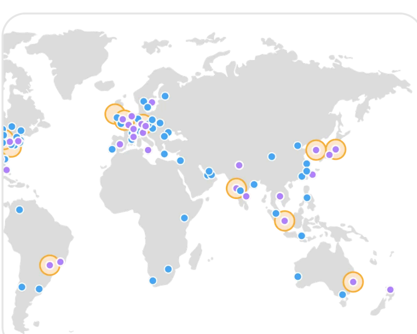
Route 53 can:

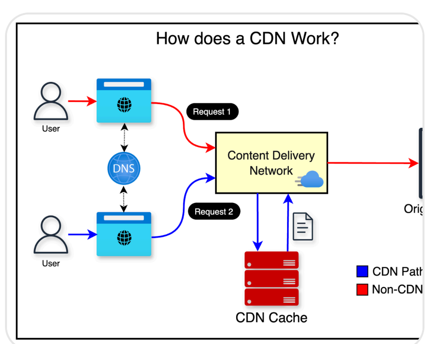
- Monitor health
- Redirect traffic to backup
- Improve availability

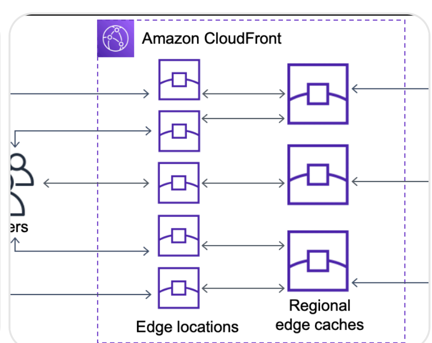
✅ Section 5 Summary

- DNS translation
- Smart routing
- Health checks
- Global performance

🚀 SECTION 6: AMAZON CLOUDFRONT (CDN)







What is a CDN?

A Content Delivery Network is a global network of cache servers that store copies of content and deliver it from nearby locations.

Amazon CloudFront Features

- Global edge locations
- Regional edge caches
- Low latency
- Secure delivery
- Pay-as-you-go pricing

Benefits

- Faster performance
- Edge security
- High scalability
- Cost efficiency
- AWS integration

Pricing Components

- Data transfer out
- HTTP/HTTPS requests
- Invalidation requests
- Dedicated IP SSL

✅ Section 6 Summary

- Global caching
- Faster content delivery
- Secure edge network

📌 FINAL MODULE TAKEAWAYS

✔ Virtual networking using VPC
✔ Secure subnet architecture
✔ Traffic routing with Route 53
✔ Fast content with CloudFront
✔ Multiple connectivity models
✔ Layered security approach

If you'd like, I can now:

✅ Turn these into **5-mark & 10-mark exam answers**
✅ Create **memory tricks & quick revision sheets**
✅ Provide **diagram drawing steps for full marks**

Just tell me 👍

Sources